How can TWINT be agile in an inert ecosystem?

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Head of Technology & Architecture

TWINT

About me

TWINT

Head of Technology & Architecture

Inpher (Startup)

5th member, doing everything
Director of Product

Swisscom

Trainee, Developer, PO
Cloud-based remote monitoring & Big Data

EPFL

Robotics & Automation

Hobbies

- Family
- Sport (Backcountry Ski, Triathlon)
- Making (3D printing, DIY, programming, electronics)

Agenda

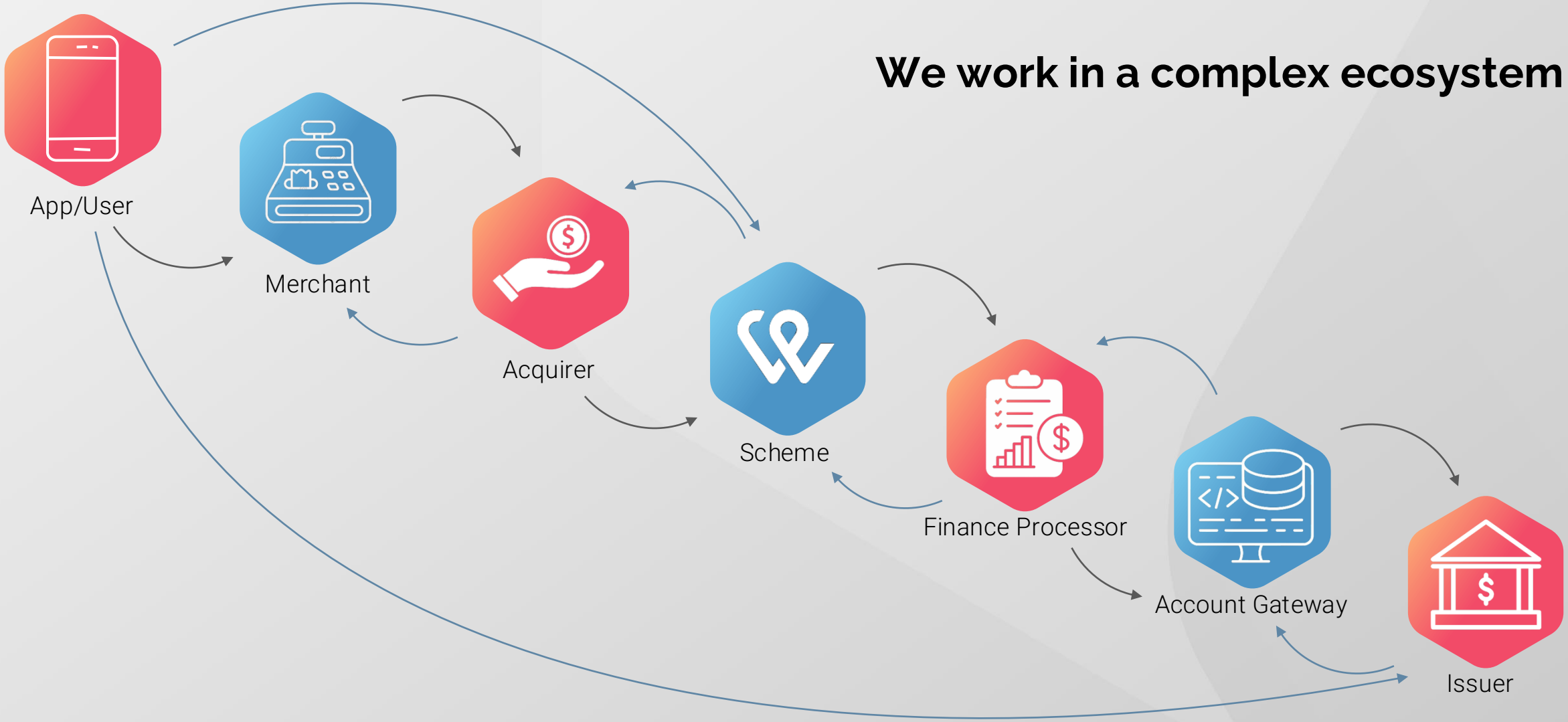
Drivers and inhibitors of agility for TWINT

How we live agility

Drivers & inhibitors of agility for TWINT

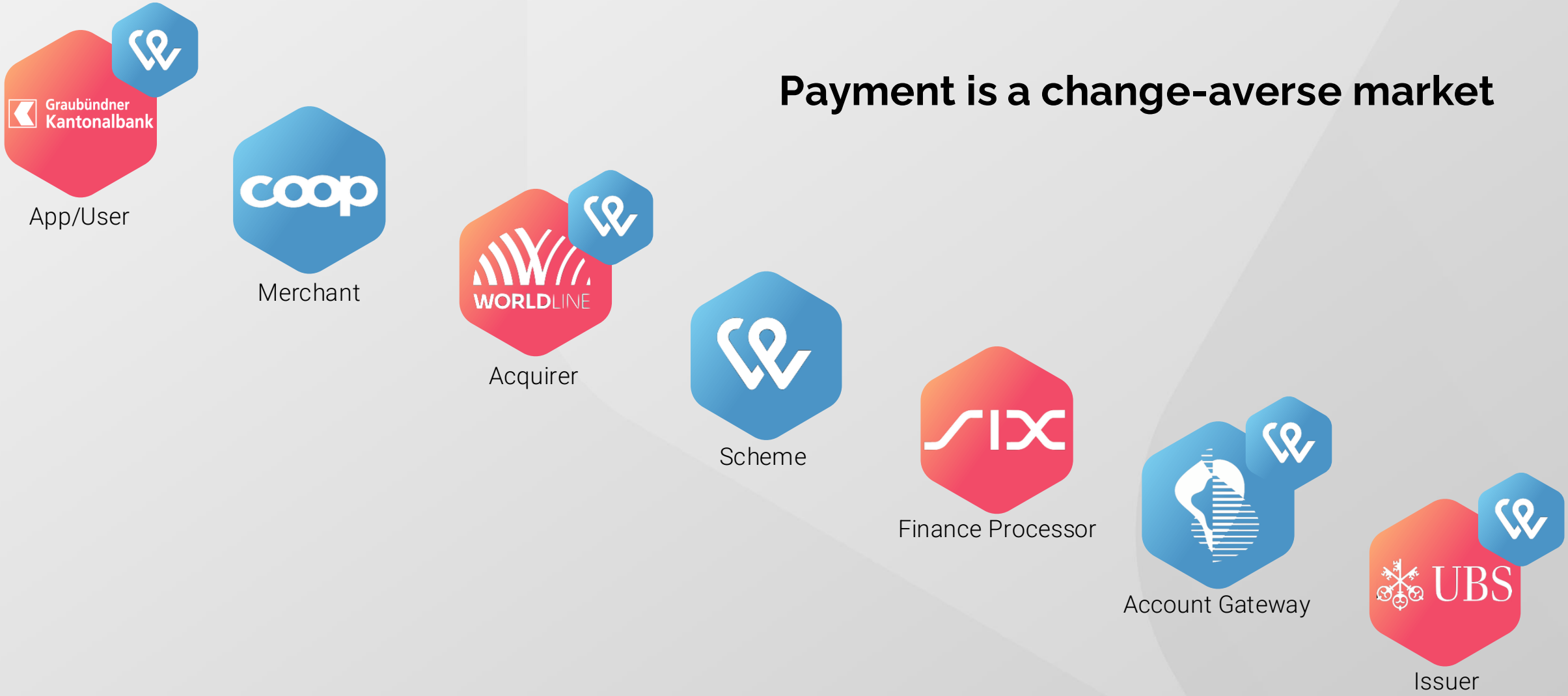
The TWINT Ecosystem

We work in a complex ecosystem



The TWINT Ecosystem

Payment is a change-averse market



We’re agile…

Drivers and inhibitors for TWINT’s agility

Inhibitors

- Complex and inert ecosystem

Drivers

Regulated Industry

On that note…

Twint schmiert mit Beschwerde gegen die Finma ab

Das Bundesverwaltungsgericht hat entschieden: Twint ist ein bewilligungspflichtiges Zahlungssystem. Die Beschwerde des Betreibers wurde abgewiesen.

4.2.25

Drivers and inhibitors for TWINT's agility

Inhibitors

- Complex and inert ecosystem
- Regulated Industry

Drivers

TWINT’s beginnings

Think back 10 years ago…

TWINT needed to be **the first** to penetrate the Swiss market and establish the leader’s position in mobile payment

When technical dept gets out of hand

Drivers and inhibitors for TWINT’s agility

Inhibitors

- Complex and inert ecosystem
- Regulated Industry
- Technical debt

Drivers

Mindset

It's not my fault
that I blame everyone else
for my mistakes.
someecards
user card

Drivers and inhibitors for TWINT’s agility

Inhibitors

- Complex and inert ecosystem
- Regulated Industry
- Technical debt
- Mindset

Drivers

Smart People

Short Feedback Loop

Roughly 5Mio monthly active users

Our Stack

when people say mainframe is a dying

Source: Brand Asset Valuator study by Ogilvy, Ingo and Scholz & Friends

Brand Recognition

Strongest brand in Switzerland

Drivers and inhibitors for TWINT’s agility

Inhibitors

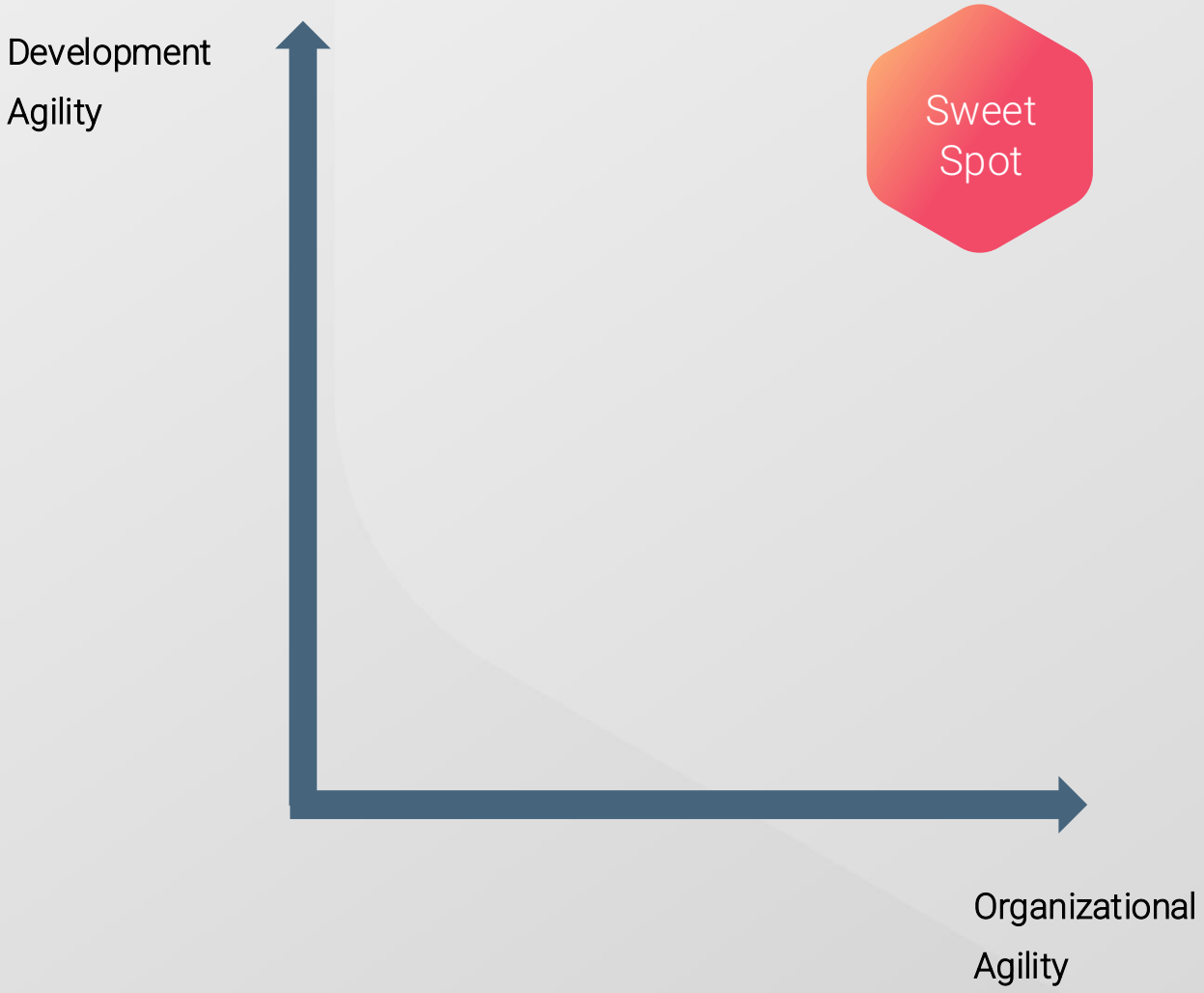
- Complex and inert ecosystem
- Regulated Industry
- Technical debt
- Mindset

Drivers

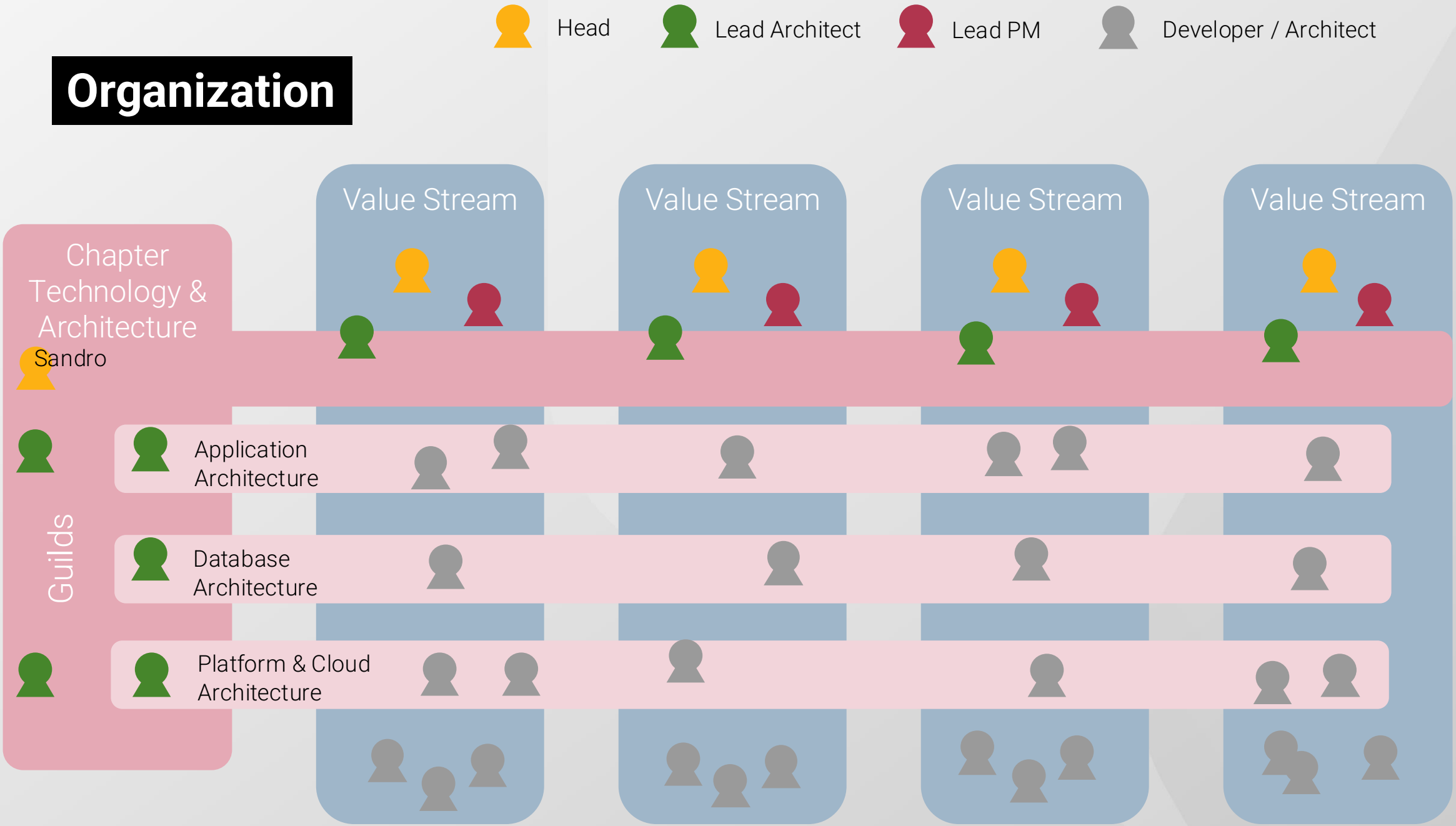
- Smart People
- Short feedback loop
- Our Stack
- Brand recognition

How can TWINT build an agile system / work in an agile way, while living in an inert ecosystem?

Agility



Organization



Organizational Principles

High Autonomy

- Value Streams are self-organized
  - Backlog, ceremonies, DoD, …
  - End to end responsible for their components

Harmoni-zation

- Scrum of architecture
  - Central source of information
  - Chapter ensures homogeneity and coordination over the entire system

Balance

- Balance control and freedom
  - Documentation first & review at early stage
  - From architecture to product

How we ensure agility

@TWINT

- Adapted Organization

Architecture F**k Ups

Architecture F**k Ups

Interface First

- Interface not generated from contract
  - We disrespected *our own* interface spec
  - Integration partner got flooded with errors and some online shops were offline

Event-driven

- Synchronous dependencies
  - Non-payment critical downstream system outage made payment crash
  - In-transaction remote calls

µ-Service

- Line concept
  - We used to have to deploy an entire TWINT system (“line”) to scale a single part
  - Deployment dependencies, stickiness, cost & time, etc…

Architecture Principles

Interface First

- Start with the contract
  - Ensure versioning
  - Ensure backward compatibility

Event-driven

- Event propagation
  - Handle business logic once and isolate core functionality from “annex”
  - Balance (eventual) consistency with nice-to-have

µ-Service

- Microservice Architecture
  - Have a good understanding of your domain and boundaries
  - CRUD use-cases and features
  - Hexagonal: Ports and adapters

How we ensure agility

@TWINT

- Adapted Organization
- Architecture Principles

Platform Services

Self-Service

- Developers can consume platform services in a self-service manner
  - Repos, Templates / Base Images, Pipeline, Vault, etc.
  - Developer has autonomy and agility in building and deploying components

Automation

- High degree of automation & coverage
  - Reduces impact of human failure and increases a developer's agility/flexibility
  - Helps in ensuring compliance (no human operator access on prod)

DevOps

- Development, Testing, Deployment and Operational responsibility with the engineer
  - Ensures confidence in a deployment
  - Empowers the developer and gives end to end responsibility

How we ensure agility

@TWINT

- Adapted Organization
- Architecture Principles
- Platform Services

Time to market

Releases

- Waterfall towards CI
  - Currently 9 releases / year, releasing “the entire system”
  - But in 5 components

Fail Fast

- Feedback and improve (or kill)
  - From Hackathon to Story Mapping, Iterations, Scope negotiations and MVP roll out -> to an evolving product

AB Testing

- We can run experiments on production
  - This helps us make informed and data-driven decisions about features or configuration

How we ensure agility

@TWINT

- Adapted Organization
- Architecture Principles
- Platform Services
- “Short” Time to market

Conclusion

Questions?

Contact

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